



Subject Areas:

applied mathematics, differential equations, statistical physics

Keywords:

early warning signals, tipping points, numerical methods, statistical inference, multiscale dynamics

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Forecasting tipping times from data

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Combining recent results in nonautonomous dynamics and statistical estimation for stochastic differential equations we derive a fully data-driven pipeline for the accurate forecast of tipping times in complex systems.

1. Introduction

We are concerned with the problem of forecasting the time at which complex systems will undergo critical transitions given past and present data. The forecast is a (linear) extrapolation of the estimated return rate of nonlinear systems slowly ramped towards a saddle-node bifurcation. In § 2 we introduce the mathematical formalism of tipping times in slow-fast stochastic processes. In § 3 we formulate the extrapolation problem and present three techniques to estimate the return rate of a given sample, two of which have already been proposed in the literature and a new one which we introduce. In § 4 we derive a series of conditions that allow for the accurate forecast of the tipping times to be considered when applying this methodology to collected measurements or experiments data. In § 5 we apply the forecasting methodology and the observation protocols of §3 and §4 to simulated and real-world timeseries data. In § 6 we conclude.

2. Tipping time of a slowly ramped systems

The prototypical case of critical transitions in complex system is that of a dynamic saddle-node normal form, with slow (linear) passage through the bifurcation and (additive) noisy fluctuations modelled as an Itô stochastic differential equation of the form

$$dX_t = f(X_t, \mu(t)) dt + \sigma dW_t, \quad f(x, \mu) := -\mu - x^2, \quad \mu(t) := \mu_0 + \varepsilon t, \quad (2.1)$$

where $X_t \in \mathbb{R}$ is the state of the system, perturbed by a Wiener process W_t with noise amplitude $\sigma > 0$, and $\mu(t)$ is the bifurcation parameter, ramped with constant speed $\varepsilon > 0$ from an initial value $\mu_0 < 0$. The drift term $f(x, \mu)$ in (2.1) has two equilibria $x_{\pm}(\mu) = \pm\sqrt{-\mu}$ for any $\mu < 0$; at $\mu = 0$ the two branches collide $x_+(0) = x_-(0) = 0$ in a saddle-node bifurcation and for $\mu > 0$ they annihilate each other. Noisy perturbations around the stable equilibrium $x_+(\mu)$ decay exponentially fast at a rate dictated by the leading eigenvalue of the linearized vector field

$$\lambda(\mu) := \partial_x f(x_+(\mu), \mu) = -2\sqrt{-\mu} < 0, \quad \mu < 0. \quad (2.2)$$

Specifically, the quantity $1/|\lambda|$ identifies the *correlation* or *relaxation time* of noisy fluctuations towards $x_+(\mu)$. Evidently, $|\lambda| \downarrow 0$ as $\mu \uparrow 0$, which in turn gives asymptotically longer relaxation time and eventually the blow-up $1/|\lambda| \rightarrow \infty$; this phenomenon is known as *critical slowing down*. This behaviour is generic for any complex system close enough to a saddle-node bifurcation as the nonlinear dynamics is reduced to the normal form (2.1) by a smooth change in coordinates.

For a given ramp speed $\varepsilon > 0$, the time at which the leading eigenvalue (2.2) vanishes corresponds to the time at which the forcing parameter $\mu(t)$ reaches the saddle-node bifurcation $\mu(t) = 0$, which is $\tau := -\mu_0/\varepsilon > 0$. Notice that one can rewrite the linear ramp of the forcing parameter to be independent of the initial condition $\mu_0 < 0$ and parametrized by the time horizon τ at which the saddle-node bifurcation is reached, i.e.

$$\mu(t) = \mu_0 + \varepsilon t = \varepsilon(t - \tau) < 0 \quad \forall t < \tau. \quad (2.3)$$

Substituting (2.3) in (2.2) gives us an explicit time dependence of the approach to the saddle-node bifurcation

$$\lambda(t) = -2\sqrt{\varepsilon(\tau - t)} \uparrow 0, \quad t \uparrow \tau. \quad (2.4)$$

(a) Deterministic tipping time

Let us first consider the deterministic limit $\sigma \downarrow 0$ for which the nonautonomous system (2.1) is known as the *dynamic* saddle-node normal form [1]. The time horizon τ (henceforth *breaking time*) identifies the loss of hyperbolicity of the stable equilibrium branch $x_+(t) := x_+(\mu(t))$ as per (2.4). In the nonautonomous regime however, the classic notion of $x_+(t)$ is replaced by the pullback attractor $x_{pb}(t)$ which is the unique solution of (2.1) (in the absence of noise $\sigma = 0$) satisfying

$$\lim_{t \rightarrow -\infty} |x_{pb}(t) - x_+(\mu(t))| = 0. \quad (2.5)$$

For a slow passage through the saddle-node bifurcation, i.e. $0 < \varepsilon \ll 1$, provided that $\mu_0 = \lim_{t \rightarrow -\infty} \mu(t) \ll 0$, the pullback attractor converges to a slow manifold [1, Thm. 2.1.7], which tracks the stable equilibrium branch $x_+(\mu(t))$. Specifically, the $x_{pb}(t)$ solution of (2.1) lags behind the drift of the *quasi-stationary equilibrium* $x_+(t)$ and persists past the breaking time τ for some time before blowing-up to $-\infty$. The trajectory of a slow passage through a saddle-node will thus undergo a bifurcation-induced critical transition at a delayed time $\tau_{pb} > \tau$ (henceforth *tipping time*), creating a window of opportunity $\Delta\tau := \tau_{pb} - \tau > 0$ to reverse the forcing ε and avoid the catastrophic dynamical saddle-node. To quantify the tipping time, and thus identify the window of opportunity, we follow the procedure outlined in [2]. We start by writing the dynamic

saddle-node normal form as a Riccati equation

$$\frac{dx}{dt} = f(x, \mu(t)) = \sum_{k=0}^2 c_k(t) x^k, \quad (2.6)$$

with $c_0(t) = \varepsilon(\tau - t)$, $c_1(t) = 0$ and $c_2(t) = -1$, and convert it into a second-order, linear system via change of variables $x \mapsto -(c_2(t)y)^{-1}(dy/dt)$

$$\frac{d^2 y}{dt^2} = c_0(t)y. \quad (2.7)$$

Eq. (2.7) becomes Airy equation in the rescaled time $s := \varepsilon^{1/3}(\tau - t)$

$$\frac{d^2 y}{ds^2} = sy, \quad (2.8)$$

whose general solution is a linear combination of Airy functions of the first $\text{Ai}(s)$ and second kind $\text{Bi}(s)$ [3, § 10.4]. By backward substitution, and in the past limit $t \rightarrow -\infty$, the pullback attractor solving (2.6) is given by

$$x_{\text{pb}}(s) = -\varepsilon^{1/3} \frac{1}{\text{Ai}(s)} \frac{d \text{Ai}}{ds}. \quad (2.9)$$

From (2.9), $x_{\text{pb}} \rightarrow -\infty$ as $\text{Ai} \rightarrow 0$, which occurs at $s \approx 2.338$. It follows trivially that the tipping time of the pullback attractor is given by $\tau_{\text{pb}} \approx \tau + 2.338 \varepsilon^{-1/3}$ and thus the opportunity window scales with the reciprocal of the cubic root of the parameter speed, i.e. $\Delta\tau \propto \varepsilon^{-1/3}$.

(b) Tipping time distribution

We now consider the more general case of stochastic fluctuations of slow passage through the bifurcation by restoring $\sigma > 0$ in (2.1). As a result, the tipping time τ_{pb} is treated as a random variable τ_{tip} with an associated probability distribution. To characterize τ_{pb} for general model of slow-fast stochastic saddle-node normal form we follow the procedures outlined in [4,5]. To start, consider again the singular limit $\varepsilon \downarrow 0$ of (2.1), written as a gradient system

$$dX_t = -V'(X_t, \mu) dt + \sqrt{2D} dW_t, \quad V(x, \mu) := \mu x + \frac{x^3}{3}, \quad (2.10)$$

where $V(x, \mu)$ is the potential of the vector field $f(x, \mu)$ and $D := \sigma^2/2$ is the diffusion intensity. The equilibria $x_{\pm}(\mu) = \pm\sqrt{-\mu}$ of f correspond to the stationary points of V , respectively its local minimum and local maximum. At any $\mu < 0$, the potential barrier, separating the local minimum $x_+(\mu)$ from the local maximum $x_-(\mu)$ is

$$\Delta V := V(x_-(\mu)) - V(x_+(\mu)) = \frac{4}{3}(-\mu)^{3/2}. \quad (2.11)$$

The critical transition of a solution of (2.1), starting in a neighbourhood of $x_+(\mu)$, can thus be reframed as the escape of an overdamped particle across the potential barrier ΔV driven by noisy fluctuations of intensity D . For fixed values of $\mu < 0$ such that $\Delta V \gg D$, these escapes are rare events described by Freidlin–Wentzell theory [6] with connections in statistical mechanics [7]. In particular, the event of a trajectory X_t of (2.10), starting near $x_+(\mu)$ and reaching $x_-(\mu)$ across the potential barrier ΔV , satisfies a so-called *large deviation principle*. The *first exit time* of such event [8, § 5.5]

$$\tau_{\text{exit}} := \inf\{t \geq 0 : X_t \leq x_-(\mu)\}, \quad (2.12)$$

is asymptotically exponentially distributed [9, Thm. 2.3]

$$\tau_{\text{exit}} \stackrel{d}{\approx} \exp(k(\mu)), \quad D \downarrow 0, \quad (2.13)$$

where the rate parameter is given by Kramers formula

$$k(\mu) = \frac{\sqrt{V'''(x_+(\mu)) |V''(x_-(\mu))|}}{2\pi} \exp\left(-\frac{\Delta V}{D}\right) = \frac{\sqrt{-\mu}}{\pi} \exp\left(-\frac{4(-\mu)^{3/2}}{3D}\right). \quad (2.14)$$

Notice that the reciprocal of the escape rate in (2.14) gives the *mean first exit time* of (2.12), i.e. $\mathbb{E}[\tau_{\text{exit}}] = 1/k(\mu)$. This result, originally formulated for stationary diffusion [10] and jump processes [11], was then extended to the dynamic saddle-node normal form (2.1). Remarkably, the distribution of the tipping time τ_{tip} for the slow passage through the bifurcation is greatly dependent by the competing effects of the noise intensity D and the ramp speed ε . In the limiting regime $D \ll \varepsilon/2$, it was shown that the escapes occur after the breaking time τ . In particular, the approximate tipping time distribution $\tau_{\text{tip}} \stackrel{d}{\approx} p_1$, derived in [4, Eq. (19)], reads

$$p_1(t) := \frac{\varepsilon^{2/3}}{\sqrt{4\pi^5 D} \text{Bi}^2(-\mu(t)\varepsilon^{-2/3}) \mathcal{J}(t)^{1/2}} \exp\left[-\frac{\varepsilon^{2/3} \text{Ai}^2(-\mu(t)\varepsilon^{-2/3})}{4\pi^2 D \text{Bi}^2(-\mu(t)\varepsilon^{-2/3}) \mathcal{J}(t)}\right], \quad (2.15)$$

with $\mathcal{J}(t) := \int_{-\infty}^t \text{Ai}^4(-\mu(s)\varepsilon^{-2/3}) ds$, has support in the interval $\tau + 1.17\varepsilon^{-1/3} < t < \tau + 3.27\varepsilon^{-1/3}$. Notice that such interval contains the tipping time of the pullback attractor $\tau_{\text{pb}} \approx \tau + 2.338\varepsilon^{-1/3}$. Remarkably, (2.15) is approximately Gaussian locally to its maximum

$$\tau_{\text{tip}} \stackrel{d}{\approx} \mathcal{N}\left(\tau_{\text{pb}}[1 + O(D/\varepsilon)], O(D\varepsilon^{-5/3})\right). \quad (2.16)$$

In the opposite regime of large noise intensity $\varepsilon/2 \ll D$, and up to $-\mu(t) \gg D^{2/3}$, it was later shown that the approximate tipping time distribution $\tau_{\text{tip}} \stackrel{d}{\approx} p_2$ has a maximum before the breaking time τ and has the form [5, Eq. (4)]

$$p_2(t) := k(\mu(t)) \exp\left[-\frac{D}{2\pi\varepsilon} \exp\left(-\frac{\Delta V(\mu(t))}{D}\right)\right], \quad (2.17)$$

where $k(\mu(t))$ is the Kramers escape rate given in (2.14) and $\Delta V(\mu(t))$ is the potential barrier given in (2.11).

3. Forecasting problem

From § 2, we summarize three important results that motivate the forecasting of tipping times:

- (i) for a slow passage through a saddle-node bifurcation the leading eigenvalue $\lambda(t)$ explicitly depends on the breaking time τ via Eq. (2.4);
- (ii) the time at which a slowly ramped solution tips τ_{pb} is delayed with respect to the breaking time τ by a factor of $\varepsilon^{-1/3}$ that depends on the speed of the forcing, thereby creating an opportunity window $\Delta\tau$;
- (iii) in the small noise regime $\sigma = \sqrt{2D} \ll \sqrt{\varepsilon}$, the delayed tipping time of a noisy solution τ_{tip} occurs almost always after the breaking time τ , as per the support of the density $p_1(t)$ in Eq. (2.15), and on average close to the deterministic tipping time, i.e. $\mathbb{E}[\tau_{\text{tip}}] \approx \tau_{\text{pb}}$ as per Eq. (2.16).

Provided a timeseries that is slowly ramped towards a saddle-node bifurcation, with noise and speed satisfying $0 < \sigma^2 \ll \varepsilon \ll 1$, then, if we accurately forecast the breaking time τ (by 1.), we can avoid the bifurcation-induced catastrophe with high probability (by 3.) by reversing the forcing ε within a time window $\Delta\tau > 0$ (by 2.). To formulate the forecasting problem of the breaking time

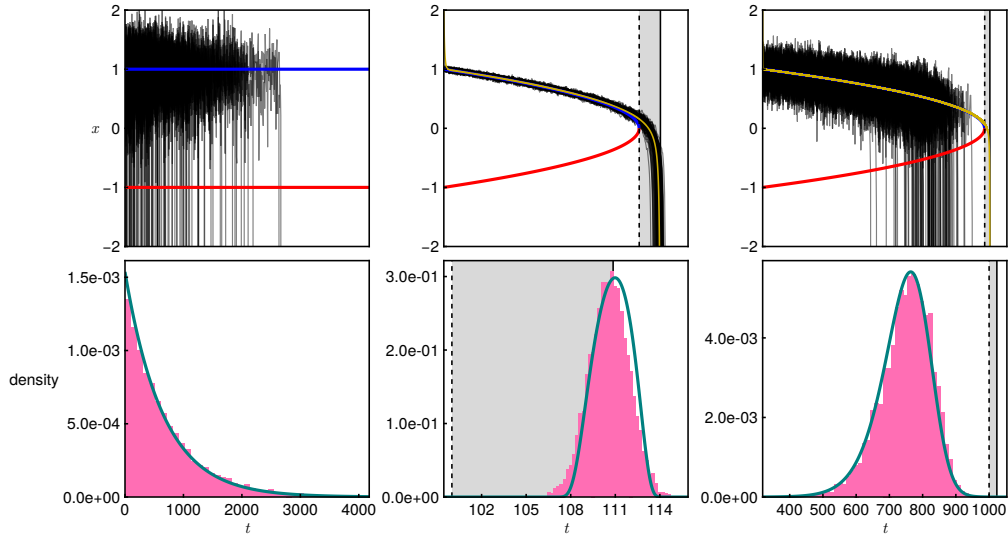


Figure 1. Analytical (teal curves, bottom row) and empirical (pink histograms, bottom row) distribution of the tipping time random variable τ_{tip} of 10,000 solutions (black curves, top row) of the saddle-node normal form (2.1) from the stable branch $x_+(\mu(t))$ (blue curves, top row) across the unstable branch $x_-(\mu(t))$ (red curves, top row). Leftmost column (N-tipping): stationary regime (2.13) with $\mu = -1$ and $D = 0.25$. Middle column (B-tipping): slow passage through the bifurcation (2.15) with $\mu_0 = -1$, $\epsilon = 10^{-2}$ and $D = 10^{-3}$. Rightmost column (N-tipping): early escapes due to large noise (2.17) with $\mu_0 = -1$, $\epsilon = 10^{-3}$ and $D = 0.06$. The window of opportunity $\Delta\tau$ (gray shaded band in the middle and rightmost columns) is bounded from below (dashed black lines) by the breaking time τ and from above (solid black lines) by the tipping time of the pullback attractor τ_{pb} (orange curve, top row).

τ we introduce the (squared) return rate $\alpha(t) := \lambda^2(t)$, where $\lambda(t)$ is given in (2.4)

$$\alpha(t) = 4\epsilon(\tau - t) > 0 \quad \forall t < \tau. \quad (3.1)$$

Notice that, while $\alpha(t) \downarrow 0$ as $t \uparrow \tau$ just like $-\lambda(t)$ does in (2.4), Eq. (3.1) establishes an affine dependence of the return rate on time t . This property benefits the forecasting of τ using timeseries data that preceede the critical transition. To understand this suppose that $\mathcal{X} := \{X_j := X_{t_j}\}_{j=1,\dots,n}$ is a solution of (2.1) with $t_j < t_{j+1}$, that is in a regime of safe distance from the bifurcation, i.e. $t_n \ll \tau$. Assuming we are able of doing so, we can estimate the true return rate (3.1) from the sample $\mathcal{X}$, thus yielding a sequence $\hat{\alpha} := \{\hat{\alpha}_j := \hat{\alpha}(t_j)\}_{j=1,\dots,n}$. Given the estimate sequence $\hat{\alpha}$, we can fit a linear model of the form $\hat{\alpha}(t) = \hat{m}t + \hat{q}$, with slope $\hat{m}$ and offset $\hat{q}$. Then, the forecasted breaking time $\hat{\tau}$ corresponds to the zero-crossing of the linear fit, which, by extrapolation, reads $\hat{\tau} = -\hat{q}/\hat{m}$. Consequentially, we translated the problem of forecasting the breaking time τ from timeseries data $\mathcal{X}$ into a trivial, linear extrapolation $\hat{\tau}$ from estimates $\hat{\alpha}$. Clearly, this problem features two bottlenecks for the accuracy of the forecasted $\hat{\tau}$: first, we need the estimates $\hat{\alpha}$ to be as close as possible to the ground truth α and; second, given those estimates $\hat{\alpha}$, we need the linear extrapolation $\hat{\tau} = -\hat{q}/\hat{m}$ to be as close as possible to the ground truth τ . In particular, depending on applications, we would prefer to slightly underestimate the breaking time $\hat{\tau} \lesssim \tau$, rather than slightly overestimating it $\hat{\tau} \gtrsim \tau$, thereby biasing our forecast to minimize missed alarms (i.e. false negatives) in favour of spurious warnings (i.e. false positives). In the remaining of this section we present three data-driven techniques to accurately extract the estimates $\hat{\alpha}$ from a sample $\mathcal{X}$ while we address the problem of precise extrapolation in §4.

(a) Ornstein-Uhlenbeck approximation

Let us start by characterizing the statistical properties of the noisy fluctuations of a solution $\mathcal{X}$ of (2.1) around the pullback attractor $x_{\text{pb}}(t)$ given in Eq. (2.9). We write the solution variable as a superposition $X_t := x_{\text{pb}}(t) + Z_t$ of a deterministic $x_{\text{pb}}(t)$ and a stochastic component Z_t . Substituting this into (2.1), recalling that by definition $dx_{\text{pb}} = f(x_{\text{pb}}, \mu)dt$, we get

$$dZ_t = (-2x_{\text{pb}}(t) Z_t - Z_t^2)dt + \sigma dW_t. \quad (3.2)$$

Let us consider time instances $t \ll \tau$ away from the saddle-node bifurcation; in this case $f(x_{\text{pb}}, \mu) \approx f(x_+(\mu), \mu)$ which implies $\partial_x f(x_{\text{pb}}, \mu) \approx \lambda(t)$. Substituting this into (3.2) gives

$$dZ_t = (\lambda(t) Z_t - Z_t^2)dt + \sigma dW_t. \quad (3.3)$$

It can be shown that $|Z_t| \approx \sigma / \sqrt{2|\lambda(t)|}$ [12, Thm. 4.1], and thus $Z_t^2 = o(|\lambda(t)Z_t|)$ for any $\sigma \ll \sqrt{\varepsilon}$ and any time $t \ll \tau$ such that $\sigma \ll \sqrt{2}(4\varepsilon(\tau - t))^{3/4}$. We can therefore reasonably approximate (3.3) by neglecting the nonlinear term Z_t^2 , thereby obtaining

$$dZ_t \approx \lambda(t) Z_t dt + \sigma dW_t. \quad (3.4)$$

The linear, nonautonomous Eq. (3.4) is a time-inhomogeneous Ornstein-Uhlenbeck process. Suppose the sample $\mathcal{X}$ is recorded across a time window $T_w > 0$ in which the underlying forcing parameter $\mu(t)$ changes very little, i.e.

$$\mathcal{X} = \{X_{t_j}\}_{j=1, \dots, n}, \quad \text{such that} \quad |\mu(t_1) - \mu(t_n)| \approx 0. \quad (3.5)$$

Then $\lambda(t) \approx \lambda$ is approximately constant and (3.4) admits a stationary solution [8, § 3.8] with zero mean $\mathbb{E}[Z_t] =: \bar{Z} = 0$ and variance $\text{Var}[Z_t] =: \gamma = \sigma^2/2|\lambda|$. Written in terms of the return rate $\alpha = \lambda^2$, the last identity reads

$$\gamma = \frac{\sigma^2}{2\sqrt{\alpha}}. \quad (3.6)$$

Let $\hat{\gamma}$ be the sample variance of $\mathcal{X}$, then (3.6) gives a sample estimate of the return rate

$$\hat{\alpha} = \frac{\sigma^4}{4\hat{\gamma}^2}. \quad (3.7)$$

The use of the sample variance as an early-warning signal dates back to the very first works in the topic and it has since been established as a benchmark precursor of critical slowing down for the detection of bifurcation-induced critical transitions (see e.g. [12] and references therein).

(b) Autoregressive approximation

In the same time window $T_w > 0$ above, the homogeneous Ornstein-Uhlenbeck process has correlation function $\text{Corr}(Y_t, Y_{t+T_w}) = \gamma \exp(-\sqrt{\alpha}T_w)$ [8, § 3.8], with γ given in (3.6). As a result, the discrete observations in sample $\mathcal{X}$ can be approximated by fitting a continuous-time autoregressive model of order one

$$X_{j+1} \approx \rho X_j + \sqrt{\gamma(1 - \rho^2)} \eta_t, \quad \eta_t \sim \mathcal{N}(0, 1), \quad (3.8)$$

where

$$\rho := \gamma^{-1} \text{Corr}(X_t, X_{t+\Delta}), \quad \Delta = t_{j+1} - t_j, \quad (3.9)$$

is the autocorrelation coefficient at lag-1. Given the sample mean $\hat{X}$ the sample autocorrelation coefficient can be obtained by maximum likelihood estimation

$$\hat{\rho} = \frac{\sum_{j=1}^{n-1} (X_j - \hat{X})(X_{j+1} - \hat{X})}{\sum_{j=1}^n (X_j - \hat{X})^2}. \quad (3.10)$$

Using (3.10) one can estimate the return rate of the sample $\mathcal{X}$ using the autoregressive approximation

$$\hat{\alpha} = \left(\frac{\log \hat{\rho}}{\Delta} \right)^2. \quad (3.11)$$

Autoregressive based early-warning signals have found widespread applications in ecology [13] and climate science [14]. In particular, estimators of the form (3.11) have been used in [15], via maximum likelihood estimation [16], for estimating the tipping time of the subpolar gyre of the Atlantic Meridional Overturning Circulation system, and in [17] for establishing a framework that quantifies the predicting skill of early-warning signals.

(c) Euler-Maruyama approximation

A solution of the nonlinear system (2.1) in a time window $T_w > 0$ as per above, can be approximated by the process [18, § 8.4]

$$X_{t+1} = X_t + f(X_t, \mu)\Delta + \sigma\sqrt{\Delta}\xi_t, \quad \xi_t \sim \mathcal{N}(0, 1). \quad (3.12)$$

The sequence above is the Euler-Maruyama approximation of (2.1) and it corresponds to the stochastic generalization of the Euler time-stepping method for ordinary differential equations. We can rearrange the terms in (3.12) to define the first-order time increments of the Euler-Maruyama approximation

$$Y_t := \frac{X_{t+1} - X_t}{\Delta} = f(X_t, \mu) + \tilde{\xi}_t, \quad \tilde{\xi}_t \sim \mathcal{N}\left(0, \frac{\sigma^2}{\Delta}\right). \quad (3.13)$$

Provided a sample $\mathcal{X} = \{X_j\}_{j=1, \dots, n}$ in a time window $T_w = n\Delta$, and a choice of the model function f , Eq. (3.13) can be interpreted as a ordinary least square regression problem, with the residuals $r_j := Y_{j+1} - f(X_j) = \tilde{\xi}_t$ being normally distributed. This technique is part of a broader class of drift inference methods called *quasi-maximum likelihood estimators* [19]. Away from the saddle-node bifurcation, the sample is approximately an Ornstein-Uhlenbeck process of the form (3.4) and thus we can reasonably fit a linear model of the form $f(x, \theta) = \theta x$ to the time increments Y_j . Substituting this ansatz into (3.13), and minimizing the square root of the sum of the squared residuals across the sample, yields the ordinary least square problem

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}} \left(\sqrt{\sum_{j=1}^{n-1} (Y_j - f(X_j, \theta))^2} \right). \quad (3.14)$$

Notice that $f \in \mathbb{P}_1[x]$, meaning that the problem (3.14) is linear in θ ; the solution of the ordinary (linear) least squares problem is thus found by the normal equation

$$\hat{\theta} = (A^T A)^{-1} A^T Y, \quad (3.15)$$

where $Y = (Y_j)_{j=1, \dots, n-1}$ is the vector of observed time increments of the sample $\mathcal{X}$ and $A = (X_j)_{j=1, \dots, n-1}$ is the model vector of the linear fit f to the sample $\mathcal{X}$. Notice that, by definition, $\hat{\theta}$ is an estimate of the leading eigenvalue λ of the underlying nonlinear evolution and thus

$$\hat{\alpha} = \hat{\theta}^2, \quad (3.16)$$

is a sample estimate of the return rate using an ordinary linear least square approximation. The application of quasi-maximum likelihood estimations for the construction of early-warning signals is much more recent than the more established autoregressive and Ornstein-Uhlenbeck approximation, with early adoptions found in [20].

4. Observation protocol

The return rate estimators using the Ornstein-Uhlenbeck (3.7), autoregressive (3.11) and linear least squares (3.16) approximations all provide a data-drive pipeline to extract $\hat{\alpha}$ from a given

sample $\mathcal{X}$ collected over a time window $T_w = n\Delta$, with $n > 0$ number of observations sampled at (regular) frequency Δ . As noted at the beginning of § 3 however, these estimates are only useful to forecast the tipping time $\hat{\tau}$ of the given sample $\mathcal{X}$. In this section we address this problem in two parts. First, we discuss the limitations and uncertainties associated in the estimators of the return rate $\hat{\alpha}$ derived above. Second, we formulate the extrapolation problem of $\hat{\tau}$ given a sequence $\hat{\alpha} = \{\hat{\alpha}_j\}_{j=1,\dots,n}$ of optimal estimators of the return rate and provide a data-driven pipeline for the accurate forecasting problem.

(a) Estimating the return rate

In addressing the problem of estimating the return rate from a finite sample we encounter three major limitations: one is associated to the discrepancy between the return rate evaluated at the quasi-steady equilibrium and the one observed from fluctuating residuals following the pullback attractor and; two are associated with *finite-sample effects* that need to be considered when designing an optimal observation protocol with the aim of forecasting tipping times.

(i) Admissible time horizon

The analytical return rate (3.1), and thus the sample estimators (3.7), (3.11) and (3.16), is obtained from the time-dependent eigenvalue (2.4), which is itself evaluated along the drifting quasi-steady equilibrium $x_+(\mu(t))$. However, a sample $\mathcal{X}$ that is a solution of (2.1) follows the pullback attractor (2.5), which lags behind $x_+(\mu(t))$ at any $t < \tau$. After a fast relaxation of the initial condition, in the slow regime the distance at which $x_{pb}(t)$ tracks $x_+(\mu(t))$ increases with time and it reaches its maximum at the saddle-node bifurcation $|x_{pb}(t) - x_+(\mu(t))| = O(\varepsilon^{1/3})$, $t \uparrow \tau$ [1, Thm. 2.2.2]. We are thus interested to quantify a time horizon $t = T_c$ for which the return rate $\alpha(t)$ of the quasi-steady equilibrium is accurate, at least to first-order, to that of the pullback attractor $\alpha_{pb}(t) = \lambda_{pb}^2(t) = 4x_{pb}^2(t)$. Sufficiently far from the saddle-node bifurcation we can exploit the asymptotic expansion of Airy's function [3, § 10.4] to approximate (2.9) to first-order

$$x_{pb}(s) \approx \varepsilon^{1/3} \left(\sqrt{s} + \frac{1}{4s} \right), \quad s = \varepsilon^{1/3}(\tau - t) \rightarrow \infty. \quad (4.1)$$

In this regime, the return rate for the pullback attractor is thus

$$\alpha_{pb}(t) = \alpha(t) + \frac{1}{4(\tau - t)^2} + 2\sqrt{\frac{\varepsilon}{\tau - t}}, \quad (4.2)$$

and dividing each term in (4.2) by $\alpha(t)$ gives

$$\frac{\alpha_{pb}(t)}{\alpha(t)} = 1 + \frac{1}{16\varepsilon(\tau - t)^3} + \frac{1}{2\sqrt{\varepsilon(\tau - t)^3}} = 1 + 2\eta(t) + \eta^2(t) = (1 + \eta(t))^2, \quad (4.3)$$

where $\eta(t) = \frac{1}{4\sqrt{\varepsilon(\tau - t)^3}}$ quantifies the lag correction of the return rate of the pullback attractor. From (4.3) it follows that $\alpha(t) \approx \alpha_{pb}(t)$ at any time $t \ll \tau$ such that $\eta(t) \ll 1$, from which we obtain the admissible time horizon

$$T_c \lesssim \tau - \varepsilon^{-1/3}. \quad (4.4)$$

The critical threshold (4.4) gives us a criterion for which the estimation of the frozen return rate yields admissible results for the extrapolation of the tipping time, as the lag correction $\eta(t)$ of the pullback attractor with respect to the quasi-steady equilibrium $x_+(\mu(t))$ is negligible.

(ii) Biased estimators

In our data-driven pipeline, the estimation of the return rate is performed out of a sample $\mathcal{X}$ that is assumed to be a solution of (2.1). Provided that the sample is in the admissible time horizon $t < T_c$ quantified above, a local estimation of $\alpha(t)$ requires that the sample $\mathcal{X}$ is recorded over a window of time in which the leading eigenvalue $\lambda(t)$ at the quasi-steady equilibrium remains relatively unchanged. This in turn requires the sample to be limited in a finite observation

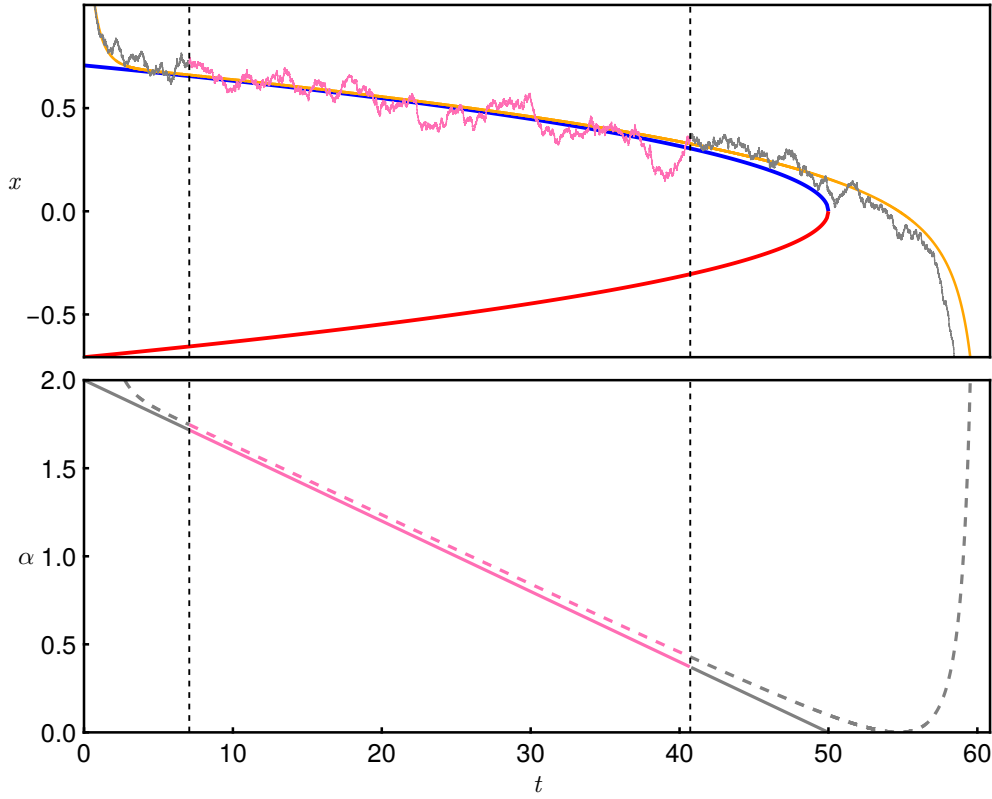


Figure 2. Accuracy of the return rate of quasi-steady equilibrium $\alpha(t)$ (solid pink curve in the bottom plot) with respect to that of the pullback attractor $\alpha_{pb}(t)$ (dashed pink curve in the bottom plot) for the saddle-node normal form (2.1). The solution $\mathcal{X}$ (pink curve in the top plot) follows the pullback attractor x_{pb} (orange curve) which track the quasi-steady equilibrium $x_+(t)$ (blue curve) with speed $\varepsilon = 10^{-2}$ and noisy fluctuations $\sigma = 0.6\sqrt{\varepsilon}$. The time window in which $\alpha(t) \approx \alpha_{pb}(t)$ to first order (dashed black lines) is given by the extinction of the transient of the pullback attractor on the lower bound (which here is taken as ten times the relaxation time from the initial condition $x_{pb}(0) = 2$) and by the admissible time horizon (4.4).

window $T_w = n\Delta$ that induces biases and uncertainties in the estimation of the return rate. We shall henceforth adopt an ensemble approach in which $\Omega = \{\mathcal{X}_j\}_{j=1,\dots,N}$ denotes a collection of $N > 1$ independent and identically distributed finite samples that correspond to different noise realizations of (2.1). Then let $\hat{\beta}$ be a generic random variable estimating some ground truth β from a single sample in the ensemble. Denote $\mathbb{E}[\hat{\beta}]$ to be the ensemble mean of $\hat{\beta}$ over Ω and $\text{Var}[\hat{\beta}]$ to be the variance of $\hat{\beta}$ over the ensemble Ω . We can thus identify the bias $b_{\hat{\beta}}$ and uncertainty $u_{\hat{\beta}}$ of the estimator $\hat{\beta}$ to be

$$b_{\hat{\beta}} := \mathbb{E}[\hat{\beta} - \beta] \quad \text{and} \quad u_{\hat{\beta}} := \sqrt{\text{Var}[\hat{\beta}]}.$$
(4.5)

Depending on the context, the bias and uncertainty of statistical estimators are also referred to as accuracy and precision respectively; here these two terminologies will be used interchangeably. In the following we shall assume each sample $\mathcal{X}_j$ in the ensemble Ω to be a sequence of stationary residuals obtained by detrending a finite windowed solution of (2.1) that is collected sufficiently away from the saddle-node bifurcation. The resulting Ornstein-Uhlenbeck process (3.4) can be approximated by an autoregressive model of order one (3.8).

Bias in the sample variance The return rate estimator (3.7) uses the sample variance in the Ornstein-Uhlenbeck approximation. It is a standard result that the sample variance $\hat{\gamma} = \text{Var}[\mathcal{X}]$ is a biased estimator for autocorrelated data. Specifically, its accuracy and precision at lag-1 are given by [21, Eq. (4.6)]

$$b_{\hat{\gamma}} = -\frac{2\gamma\rho}{(n-1)(1-\rho)} \quad \text{and} \quad u_{\hat{\gamma}} = \sqrt{\frac{2\gamma^2(1+\rho^2)}{(n-1)(1-\rho^2)}}, \quad (4.6)$$

where ρ is the autocorrelation coefficient given in (3.9). Following a standard delta method [22, § 5.5.4], and using (3.6), we obtain a second-order approximation of the bias in the estimated leading eigenvalue within the Ornstein-Uhlenbeck assumption, which reads

$$b_{\hat{\lambda}}^{\text{OUP}} = \frac{|\lambda|}{\gamma} b_{\hat{\gamma}} - \frac{|\lambda|}{\gamma^2} u_{\hat{\gamma}}^2. \quad (4.7)$$

Plugging (4.6) into (4.7) yields a closed form expression for the bias in the estimation of the leading eigenvalue of the sample $\mathcal{X}$ using the variance of its residuals

$$b_{\hat{\lambda}}^{\text{OUP}} = -\frac{2|\lambda|(1+\rho+2\rho^2)}{(n-1)(1-\rho^2)}. \quad (4.8)$$

Bias in the sample autocorrelation The return rate estimator (3.11) uses the squared logarithm of the sample autocorrelation (3.10), which is quantified in [23, Eq. (21)] to be

$$b_{\hat{\rho}} = -\frac{1+3\rho}{n-1}.$$

Applying again the delta method, the above yields the bias in the leading eigenvalue [24, Eq. (12)]

$$b_{\hat{\lambda}}^{\text{ACI}} = -\frac{3+\rho^{-2}}{2\Delta(n-1)}. \quad (4.9)$$

Bias in the ordinary least squares solution The return rate estimator (3.16) is the square the ordinary least squares solution of the Euler-Maruyama approximation (3.15), which coincides with the estimated leading eigenvalue of the sample $\mathcal{X}$ by definition. Its bias is thus more straightforward to compute [25, Eq. (4.5)] and it reads

$$b_{\hat{\lambda}}^{\text{OLS}} = \frac{\Delta\lambda^2}{2} - \frac{1+3\rho}{\Delta(n-1)}. \quad (4.10)$$

Notice that, while (4.8) and (4.9) are negative for all samples up to the saddle-node bifurcation $\lambda = -2\sqrt{-\mu} < 0$, the bias in the least squares estimator (4.10) changes sign at

$$\mu = -\frac{1+3\rho}{2\Delta^2(n-1)}, \quad (4.11)$$

for which (3.16) is thus an unbiased estimator of the leading eigenvalue.

Accuracy and precision of the estimators Eqs. (4.8)-(4.10) measure the accuracy for the estimation of the leading eigenvalue λ of a sample. To convert those into biases of the return rate α , notice that

$$\mathbb{E}[\hat{\alpha}] = \mathbb{E}[\hat{\lambda}^2] = \mathbb{E}[\hat{\lambda}]^2 + \text{Var}[\hat{\lambda}] = (b_{\hat{\lambda}} + \lambda)^2 + u_{\hat{\lambda}}^2, \quad (4.12)$$

where in the last equality we used the definitions (4.5). Expanding the quadratic term and using the definition of the return rate α we obtain

$$b_{\hat{\alpha}} = b_{\hat{\lambda}}^2 + 2\lambda b_{\hat{\lambda}} + u_{\hat{\lambda}}^2, \quad (4.13)$$

from which we conclude that the accuracy of the estimators depend on both the accuracy $b_{\hat{\lambda}}$ and the precision $u_{\hat{\lambda}}$ of the leading eigenvalue λ . In the high sampling frequency limit the uncertainty

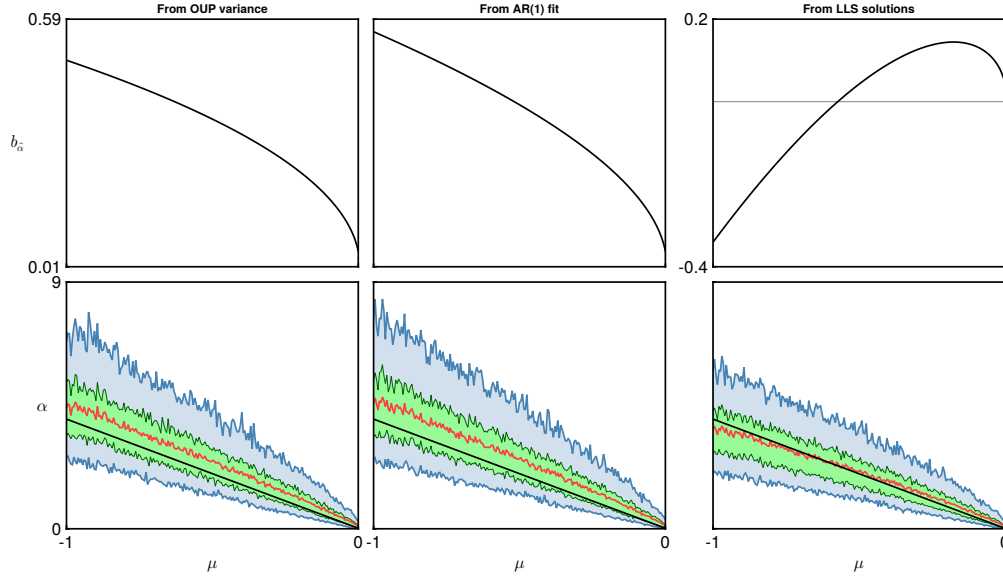


Figure 3. Comparison of the bias of the return rate estimators (4.13) (top row) obtained from the sample variance (3.7) (left), sample autocorrelation (3.11) (middle) and linear least squares solution (3.16) (right). Notice how the estimator from the sample variance and autocorrelation are positively biased up to the saddle-node bifurcation, compared to the estimator for the least squares regression which crosses zero (gray horizontal line in the top right) at the parameter value prescribed by (4.11). These results are validated by Monte Carlo simulations (bottom row) in which an ensemble Ω of $N = 200$ particles' trajectories of (2.1) are solved at fixed values parameter of μ in a discretized sweep of 200 points in $[-1, 0)$. Each trajectory forms a sample $\mathcal{X} \in \Omega$ of $n = 400$ observations of noise intensity $\sigma = 0.02$. The estimators $\hat{\alpha}$ in (3.7), (3.11) and (3.16) are then computed from the sample $\mathcal{X}$ using the prescribed methodology and the ensemble statistics, such as mean (red curve), interquartile range (green shaded region bounded by dark green curves) and 5th-95th interval (blue shaded region bounded by dark blue curves), are compared with the ground truth α .

is approximated by the Cramer–Rao inequality [22, Thm. 7.3.9]

$$u_{\lambda}^2 \rightarrow \frac{2|\lambda|}{\Delta(n-1)}, \quad \Delta \rightarrow 0. \quad (4.14)$$

(iii) Optimal observation length

The topic of selecting an optimal window length to forecast critical transitions was addressed in [26]. In particular, the authors showed that windowed observations with less than 10% of the data from a given sample yield spurious early-warning signals. Such study however only considers fixed window lengths across a solution approaching a critical transition, which neglects the changing characteristic of the solution in the slow passage through the saddle-node. Here we derive the optimal, time-dependent window length of a sample for the accurate forecasting of the tipping time. In particular, given a sample $\mathcal{X} = \{X_{t_1}, \dots, X_{t_n}\}$ recorded across a window $T_w = t_n - t_1$, we look for bounds of the form

$$L(t) \ll T_w \ll U(t), \quad (4.15)$$

in which the residuals retain the necessary properties for the estimation of the frozen return rate (3.1). The lower bound $L(t)$ is derived by imposing that the sample $\mathcal{X}$ is observed on a window

that is much longer than the relaxation time $1/|\lambda|$ which, from (2.4), reads

$$L(t) = \frac{1}{2\sqrt{\varepsilon(\tau - t)}} \ll T_w. \quad (4.16)$$

This requirement ensures that the realizations X_j in the sample are independent fluctuations around the stable equilibrium $x_+(\mu(t))$. On the other hand, the upper bound $U(t)$ is derived by imposing that the sample $\mathcal{X}$ is quasi-stationary. To derive $U(t)$ observe that, from (2.4), the instantaneous rate of change of the eigenvalue is

$$\frac{d\lambda}{dt} =: \dot{\lambda}(t) = \frac{2\varepsilon}{|\lambda(t)|},$$

which is positive and monotonically increasing for any $t < \tau$, which implies that $\dot{\lambda}(t_n) = \sup_{t_1 < t < t_n} \dot{\lambda}(t)$ over any finite time window (t_1, t_n) . By setting $T_w = t_n - t_1$ it follows

$$0 < \Delta\lambda = \lambda(t_n) - \lambda(t_1) = \int_{t_1}^{t_n} \dot{\lambda}(t) dt < \dot{\lambda}(t_n) T_w = \frac{2\varepsilon}{|\lambda(t_n)|} T_w,$$

and dividing each term by $|\lambda(t_n)|$ yields

$$\frac{\Delta\lambda}{|\lambda(t_n)|} < \frac{2\varepsilon}{\lambda^2(t_n)} T_w.$$

Our requirement for the sample $\mathcal{X}$ to be quasi-stationarity across T_w corresponds to small changes of the leading eigenvalue compared to its endpoint value, i.e. $\Delta\lambda/|\lambda(t_n)| \ll 1$, which is guaranteed by imposing

$$\frac{2\varepsilon}{\lambda^2(t_n)} T_w \ll 1,$$

yielding a sufficient condition for the upper bound for the window length

$$T_w \ll U(t) = \frac{\lambda^2(t)}{2\varepsilon} = 2(\tau - t), \quad (4.17)$$

where the last equality follows from (3.1). Plugging (4.16) and (4.17) into (4.15) provides us with the window length protocol

$$\frac{1}{2\sqrt{\varepsilon(\tau - t)}} \ll T_w \ll 2(\tau - t), \quad (4.18)$$

where notice that only the lower bound depends on the rate of change ε of the parameter.

(b) Extrapolating the tipping time

Having derived the design conditions for the collection of samples that yield optimal estimators $\hat{\alpha}$ we are now ready to specify a data-driven pipeline for the forecasting of the tipping time. As anticipated in § 3, given a sequence of estimators $\hat{\alpha} = \{\hat{\alpha}_j\}_{j=1, \dots, N}$, the forecasting of the tipping time τ_{tip} is translated into an extrapolation problem for the breaking time $\hat{\tau} = -\hat{q}/\hat{m}$ via the linear fit $\hat{\alpha}(t) = \hat{m}t + \hat{q}$. We are thus concerned in formulating the optimal setup for estimating the slope $\hat{m}$ and intercept $\hat{q}$ of the underlying return rate $\alpha(t)$ given a sample of its estimation $\hat{\alpha}(t)$. As for the Euler–Maruyama approximation outlined in § 3(c), the linear regression for the sample $\hat{\alpha}$ corresponds to a linear least squares problem whose solution is given by the normal equation

$$\begin{pmatrix} \hat{q} \\ \hat{m} \end{pmatrix} = (B^T B)^{-1} B^T \hat{\alpha}, \quad (4.19)$$

where $B = (1, t_j)_{j=1, \dots, N}$ is the model matrix and $\hat{\alpha}_j = \hat{\alpha}(t_j)$ is the estimated return rate at time t_j . The ordinary least square regression (4.19) requires that the uncertainty in the input data is homogeneous (i.e. *homoskedasticity*) to be an efficient estimator of $(\hat{q}, \hat{m})^T$. This however is clearly not the case for the return rate sample $\hat{\alpha}$ as evidenced in Figure 3; in particular, all three estimators (3.7), (3.11) and (3.16) appear to have decreasing uncertainty the closer the system is to the saddle-node bifurcation. Therefore the heterogeneity in the uncertainty of the sample $\hat{\alpha}(t)$ (i.e.

heteroskedasticity) makes the ordinary least square regression (4.19) an inefficient point estimate of the linear trend $\hat{m}$ and intercept $\hat{q}$ [27]. Assuming that, albeit heterogeneous, the uncertainty $u_{\hat{\alpha}_j}$ in the estimate $\hat{\alpha}_j$ is uncorrelated with that of any other estimate $\hat{\alpha}_k$ in the sequence $\hat{\alpha}$, then the weighted least square formulation

$$\begin{pmatrix} \hat{q} \\ \hat{m} \end{pmatrix} = (B^T W B)^{-1} B^T W \hat{\alpha}, \quad W = \text{diag}(u_{\hat{\alpha}_j}^{-2})_{j=1, \dots, N}, \quad (4.20)$$

provides the best possible (unbiased) estimator for the linear fit [28]. Using the squared residuals of the ordinary least squares solution (4.19) as heteroskedasticity-consistent estimators of the entries of the weight matrix W [27, Thm. 1], one can thus implement a fully data-driven (weighted) extrapolation problem (4.20), from the sequence $\hat{\alpha}$, to forecast $\hat{\tau}$.

5. Practical applications

6. Conclusions

Data Accessibility. The code used to generate the figures is available at the repository https://github.com/papadeiv/Research/tree/master/2027_proc_r_soc_a.

Authors' Contributions. D.P.: Conceptualization, Methodology, Formal analysis, Investigation, Writing - Original draft preparation; L.S.: Supervision, Conceptualization, Validation, Writing - Review & editing; G.D.: Project administration, Supervision, Funding acquisition, Conceptualization, Validation, Writing - Review & editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interests. We declare we have no competing interests.

Funding. D.P. and G.D. have received support from the Royal Society of New Zealand, Te Apārangi by the Marsden fund n.UOA2223; L.S. has received support from the Royal Society of New Zealand, Te Apārangi by the Marsden fund n.23-UOA-152

Appendix A

(a) Something important

Example table.

Table 1. An Example of a Table

date	Dutch policy	date	European policy
1988	Memorandum Prevention	1985	European Directive (85/339)
1991–1997	Packaging Covenant I		
1994	Law Environmental Management	1994	European Directive (94/62)
1997	Agreement Packaging and Packaging Waste		

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